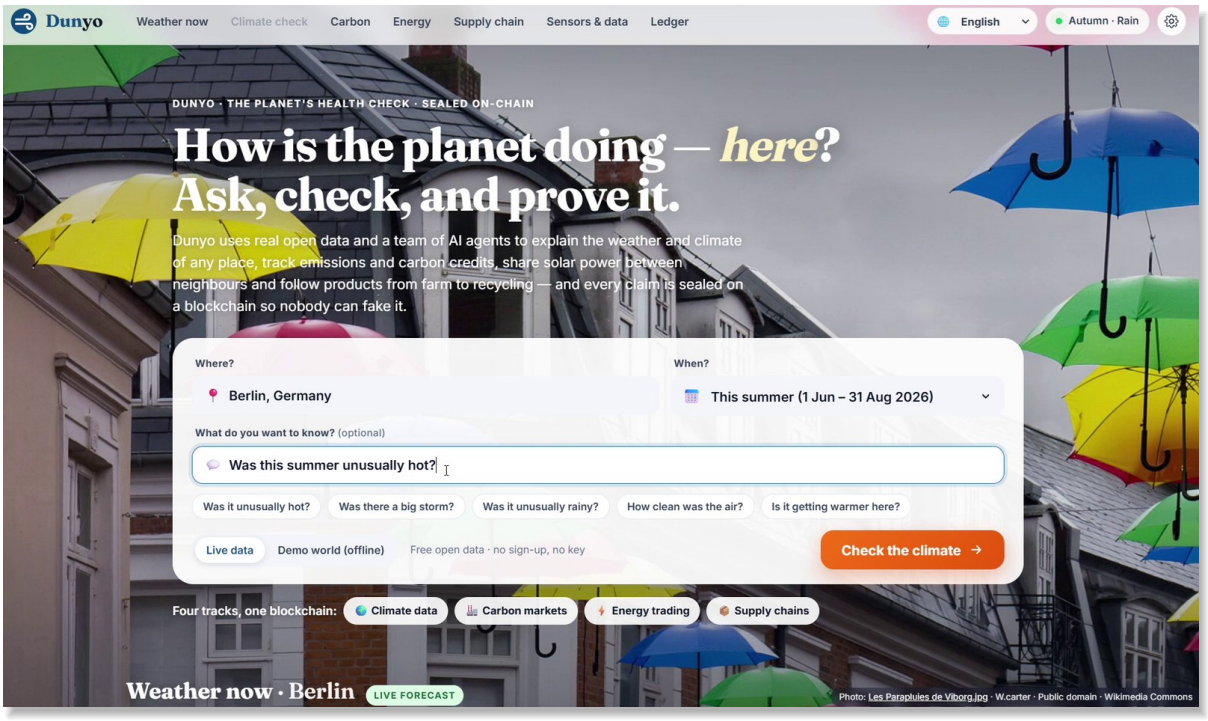
Dunyo

The planet's health check — sealed on-chain

Real open data, AI that doubts itself and one shared blockchain for climate data, carbon credits, energy trading and product passports.

1 · Carbon markets | 2 · Energy trading | 3 · Supply chains | 4 · Climate data (primary)



4	14	13	0
tracks, one ledger	open-data connectors	languages	API keys needed

Technical report
IEEE ClimateChain Global Hackathon 2026

Team Dunyo
github.com/soroorgh93/dunyo

25 October 2026
Open source · MIT licence

Executive summary

The problem

Climate action runs on *claims*: "this summer was exceptional", "our plant emitted 20,000 t", "these credits are real", "this street is solar-powered", "this product is low-carbon", "this sensor says the air is clean". The data behind them is fragmented across agencies and companies, hard for non-experts to read, and easy to misreport or count twice.

What Dunyo does

Dunyo checks each claim against real open data with AI, explains the result in plain words for anyone, and seals every input and result on *ClimateChain*, a ledger with real Ed25519 signatures, Merkle roots and smart-contract rules that block double counting. One page covers all four ClimateChain tracks, any place on Earth, in 13 languages, with no account and no API key.

Table 1: Results of a live session for Berlin on 25 October 2026 (real data, unedited).

Climate check	"Was this summer unusually hot?" → **Yes — exceptional**: heat wave 24–30 June, peak 39.7 °C, rarer than once in 1,000 years by the 1991–2020 standard, 97 % confidence; very hot days rose from 22 to 38 a year (1991–2000 vs 2016–2025).
Track 1	Boxberg power station (Climate TRACE, 12.5 Mt CO_2e/yr): 16,948 t of under-reported CO_2 caught in one week; a 5 MW solar claim of 102 credits cut to **78 verified credits** by real irradiance and physics.
Track 2	9 buildings, 32 kW of rooftop PV: **44 kWh** expected tomorrow (range 37–53 kWh), 14 kWh traded between neighbours at real day-ahead prices, €2.42 saved, 59 hourly certificates.
Track 3	Barilla Spaghetti N°5 (real Open Food Facts record): 8 signed lifecycle steps, 1.06 kg CO_2e, deposit refund and recycled-material credits — once.
Track 4	12 sensors cross-checked, 4 unreliable ones excluded, verified $PM_{2.5}$ 16.7 µg/m^3; a research lab bought the dataset and $16.00 (80 %) went to trusted sensor owners.
Ledger	All **197 blocks / 196 transactions** re-verified from scratch; changing one old number made verification fail. Every "cheat" attempt (double issuance, double retirement, wrong-hour certificate, impossible kWh, double refund, fake shipment, sensor replay) was stopped.

Why it matters. Each track answers a real market failure that regulators are now acting on: double counting in voluntary carbon markets [1], annual matching that hides fossil hours behind "100 % renewable" claims [2], digital product passports required by the EU from 2027 [3, 4], and low-cost air sensors whose readings drift in humid air [5]. Dunyo shows one coherent answer: *AI and physics check whether a claim is true; the blockchain guarantees it is unique, final and tamper-evident.*

1 The user journey

A city planner, a journalist or a student opens one page, types a place ("Berl…" → Berlin, Germany) and asks a question in plain words. The background is a real, openly licensed photograph that matches the season and today's weather. Live weather (hourly, daily, monthly outlook), air quality and a map appear immediately; meanwhile nine AI agents investigate. The answer reads like a news story (Figure 1): a direct yes/no, the key numbers, a "how unusual was it?" meter that rates exactly what was asked, and a "What you're seeing" sentence under every chart.

Can you trust it?. Three independent safeguards are shown to the user (Figure 2): two weather records (Copernicus ERA5 and NASA POWER) must agree; a critic agent tries to explain the anomaly away; and every input plus the report are hashed into a Merkle tree whose root is sealed on the ledger. *Check the seal* recomputes it; *Try to tamper* changes one number and the seal visibly breaks. Follow-up questions are answered by a local LLM (Ollama) and every number in the answer is checked against the data.

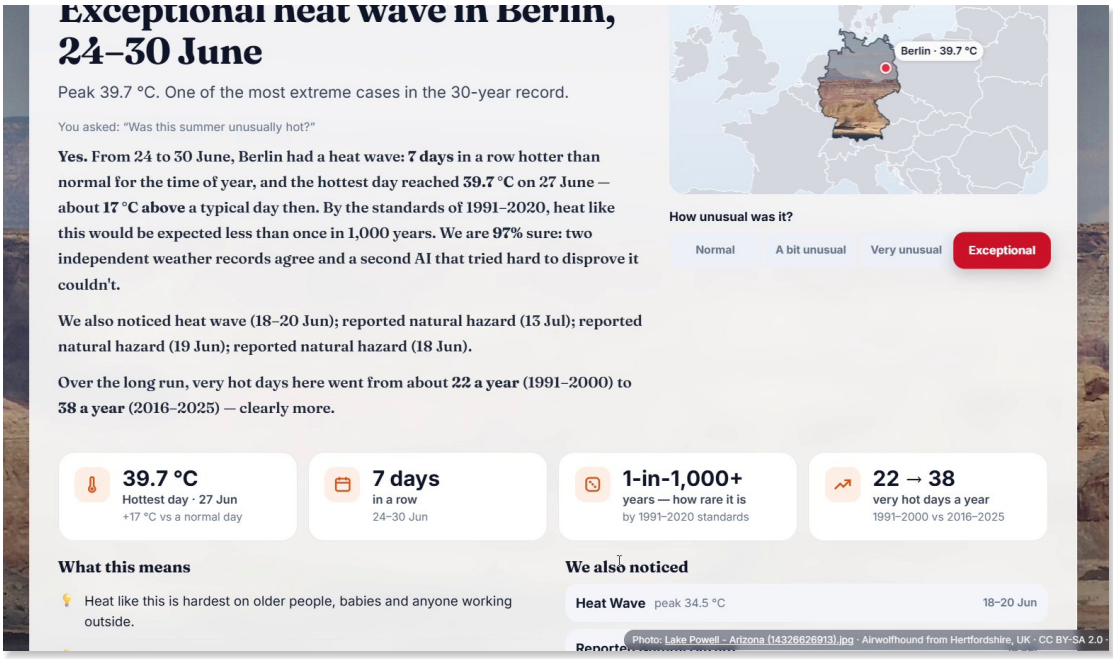


Figure 1: The answer for Berlin, summer 2026: a direct verdict, key facts and the "how unusual" meter.

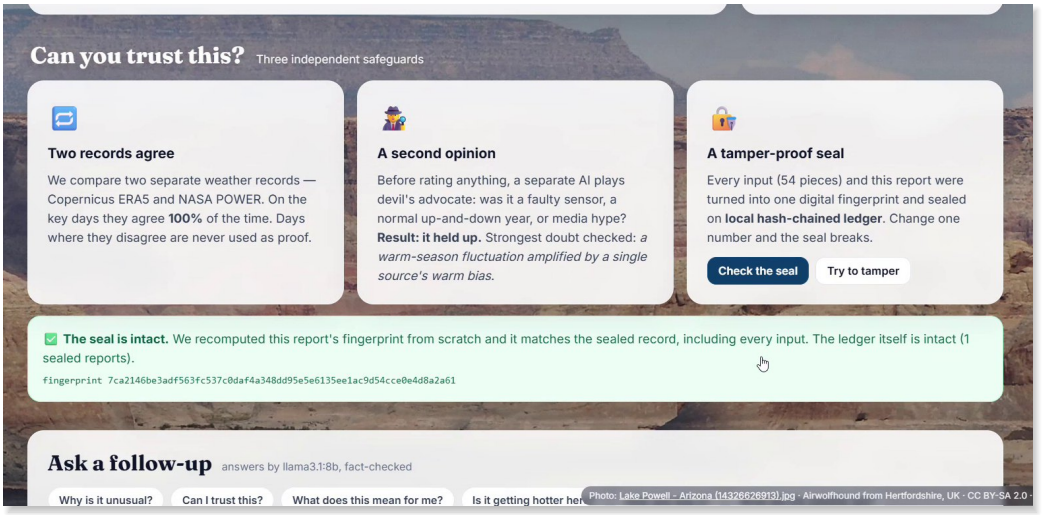


Figure 2: Trust panel: two records agree, a second AI tried to disprove the result, and the tamper test.

2 Architecture

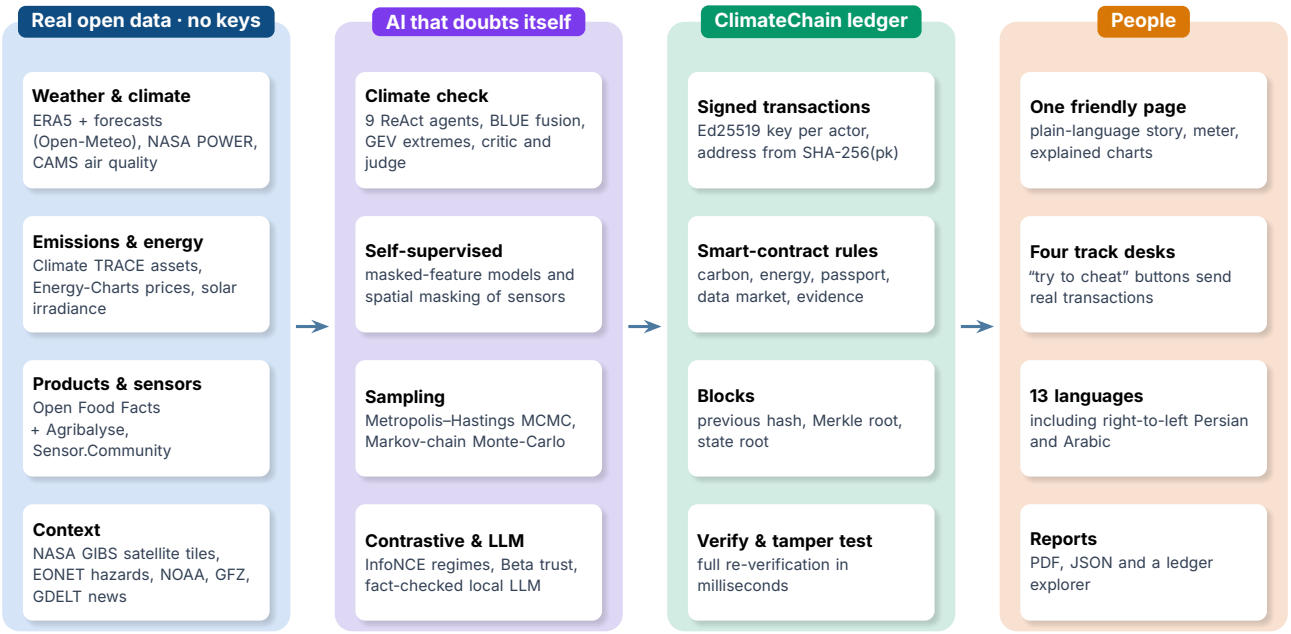


Figure 3: Data flows left to right: open data is checked by AI, every claim becomes a signed transaction, and people see the result in plain words.

Dunyo is a single Python package with a standard-library HTTP server (server-sent events stream the agents' progress) and a dependency-free web front end (hand-built SVG charts, d3-geo maps). Every connector uses short timeouts, a circuit breaker and a content-addressed cache, and every desk falls back to a clearly labelled offline demo world, so a demo never breaks. Each track runs end-to-end in 1–7 s with live data.

3 The four ClimateChain tracks

Each desk follows the same pattern: *real data → AI check → signed transactions under a contract rule → "try to cheat" buttons* that send real transactions which the contract or the AI must stop. Failed attempts stay on the chain as evidence.

Table 2: What each desk does and which attacks it blocks.

Track	Real data	AI inside	Blocked or flagged
1 Carbon	Climate TRACE facility; hourly irradiance	masked model on plant sensors; MCMC emissions; Monte-Carlo PV physics	double issuance, cross-registry crediting, double retirement, selling retired credits
2 Energy	weather forecast; Energy-Charts prices	Markov-chain cloud scenarios; load shifting; P2P clearing	wrong-hour claims, double minting, physically impossible kWh
3 Supply	Open Food Facts + Agribalyse; EU battery fields	Markov chain of lifecycle paths; shipment physics; signed label claims	double refund, wrong signer, out-of-order steps, "teleport" shipments
4 Climate data	Sensor.Community; CAMS; 30-day air + weather	spatial masking + Beta trust; contrastive regimes	replayed batches, impostor sensors; untrusted sensors earn nothing

3.1 Track 1 — Carbon markets & emissions transparency

The desk finds the nearest real emitting facility in Climate TRACE [8] (for Berlin: Boxberg power station, 12.5 Mt CO_2e a year). A 14-day plant sensor stream (stack CEMS, fuel meter, power output, ambient temperature) is generated around its real annual emissions, with two days of fuel under-reported by 30 %. A

self-supervised masked model, trained on the first week only and without labels, predicts each signal from the others; residuals above 4σ flag the tampering. A Metropolis–Hastings sampler combines the three imperfect meters into the true weekly total with a 90 % interval (234,488 t; 224,979–244,689 t) against 217,540 t reported. Emission factors follow the IPCC defaults (natural gas 56.1, bituminous coal 94.6 t CO_2/TJ) [6].

A 5 MW solar farm then claims credits for the last 30 days. Real hourly irradiance and temperature drive 400 Monte-Carlo simulations of panel performance (PR 0.76–0.86); only the conservative P10 is credited at the national grid factor [7], so 78 of 102 claimed credits are issued, traded and retired with a public certificate. Uniqueness keys on project, period and location prevent issuing the same tonne twice — also across registries — in line with the ICVCM definition of double issuance and double use [1].

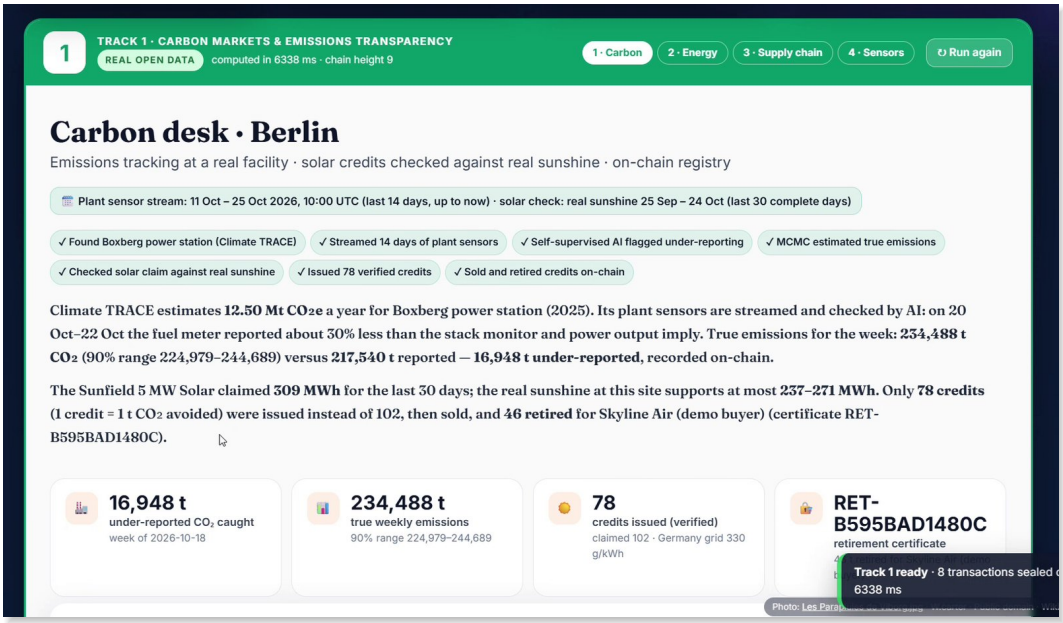


Figure 4: Track 1 desk for Berlin: emissions caught, credits verified, every step on-chain.

3.2 Track 2 — Renewable energy & energy trading

A neighbourhood of nine buildings with 32 kW of rooftop PV and a 60 kWh community battery plans tomorrow. A three-state Markov chain of cloud regimes learnt from the last 30 days samples 300 scenarios around the real forecast, giving a range instead of a single guess. Flexible loads (car charging, washing machines) move into sunny hours and earn GREEN tokens; neighbours trade hour by hour at the midpoint of retail and feed-in prices, using real day-ahead prices where Energy-Charts publishes them [9]. Every solar kWh becomes an hourly certificate: a noon certificate cannot cover an 8 pm load, the hourly-matching principle of EnergyTag [2].

3.3 Track 3 — Sustainable supply chains

Any barcode becomes a digital product passport. The product, its origins, factories, packaging and the Agribalyse life-cycle CO_2 of its food category (ADEME, via Open Food Facts [10]) are read live; when a brand publishes no origin or factory, the passport says so instead of inventing a route — exactly the gap the EU Ecodesign Regulation's Digital Product Passport will close [4]. Each lifecycle step must be signed by the role allowed to sign it, in an allowed order, hash-linked to the previous step; label claims need an accredited certifier's signature; returning the product refunds the deposit and mints recycled-material credits exactly once. An EV-battery template carries the EU Battery Regulation fields (passport from 18 February 2027; recycled-content targets for 2031 and 2036) [3].

3.4 Track 4 — Climate data & environmental monitoring (primary track)

Besides the climate check, this desk verifies citizen air sensors. Each sensor is hidden in turn and predicted from its neighbours (inverse-distance, self-supervised); agreement with the neighbours and with the Copernicus CAMS model raises a Beta-reputation trust score, large misses and humidity above 85 % (which inflates optical PM readings) lower it [14]. A contrastive encoder trained with the InfoNCE loss [12, 13] embeds 24-hour

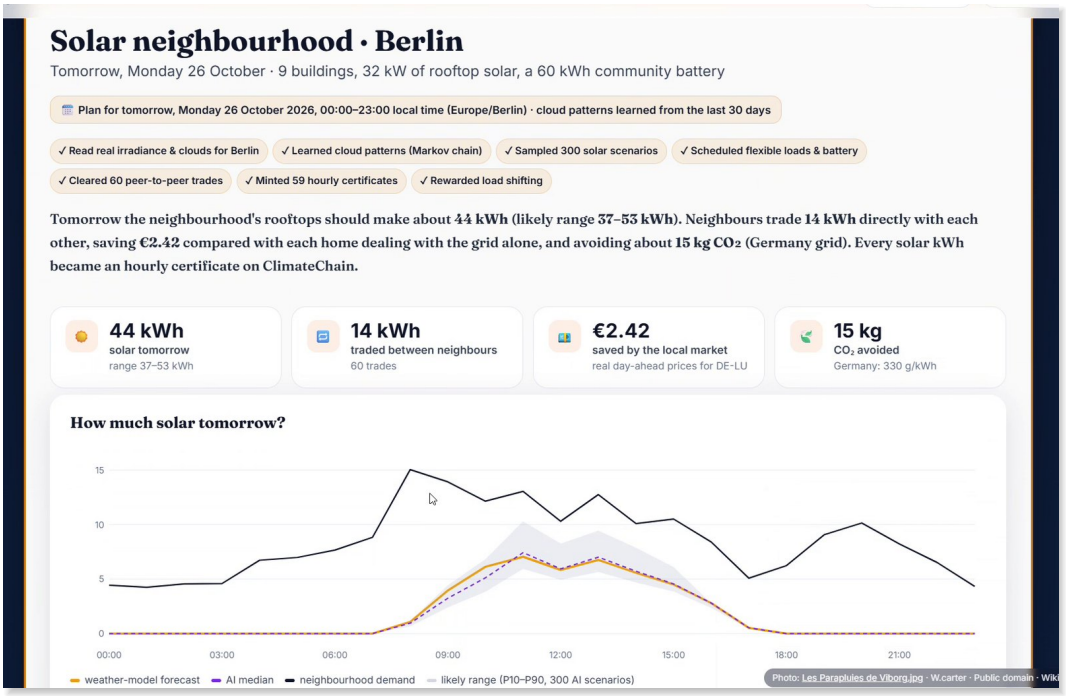


Figure 5: Track 2: tomorrow's solar range, peer-to-peer trades and hourly certificates for Berlin.

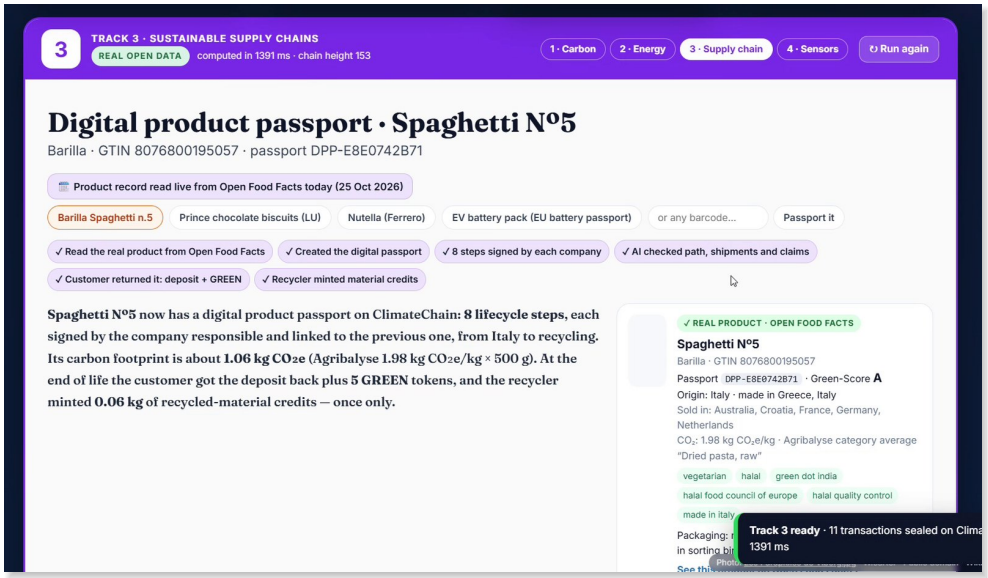


Figure 6: Track 3: a real product passport (Barilla Spaghetti Nº5) with signed steps and AI checks.

windows of the last 30 days of $PM_{2.5}$, PM_{10}, NO_2, ozone, temperature, wind and humidity; clustering the embeddings yields air-pollution regimes and the most unusual stretch. Verified datasets are listed on-chain; a buyer receives an access token and 80 % of the price is shared among sensors with trust ≥ 0.5.

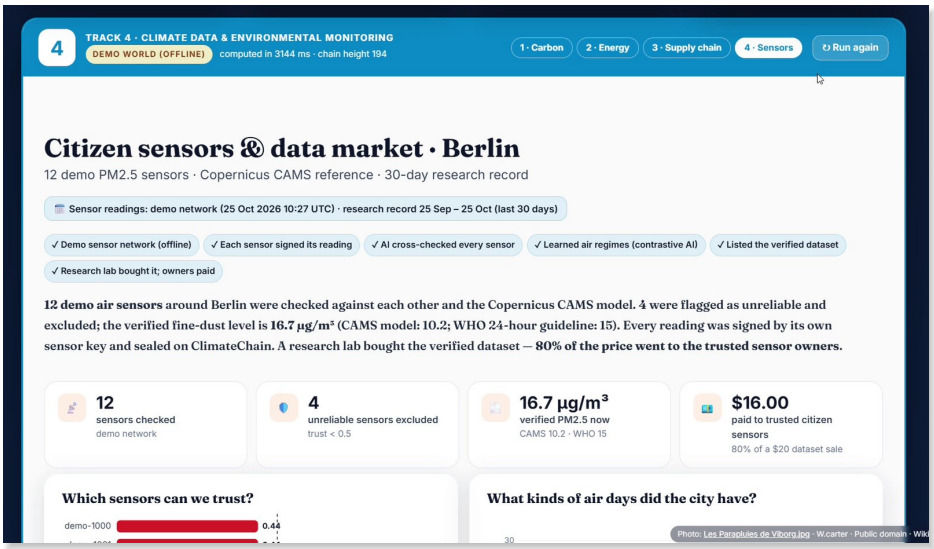


Figure 7: Track 4: sensor trust scores, contrastive air regimes and a fair data market.

4 ClimateChain

Transactions. Every actor — registry, verifier, solar farm, prosumer, factory, recycler, citizen sensor, AI validator — owns an Ed25519 key pair [17]. A transaction carries the sender, a nonce, the contract, method and arguments, and is signed over a canonical JSON encoding. The node verifies the signature and nonce, snapshots the state, runs the contract rule and either commits or reverts; reverted transactions are still recorded with their reason.

Blocks. Each block stores the previous block hash, the Merkle root [18] of its transactions and a hash of the full contract state. `verify()` recomputes every signature, hash link, Merkle root and state root from scratch; `tamper_test()` changes one old number in a copy of the chain and shows where verification fails.

Contracts. Five rule sets run natively (Python) and ship as Solidity for any EVM chain: Carbon Registry, Energy Market, Product Passport, Data Market and Climate Evidence Registry (`contracts/*.sol`). Climate-check reports are additionally anchored through the evidence registry, optionally on Sepolia via web3.py.

Why a blockchain. Climate claims must be *unique* (one tonne, one credit; one kWh, one hour), *final* (a retired credit cannot be sold) and *tamper-evident* (a report cannot be quietly edited). These are exactly the properties a shared ledger provides; truthfulness is the job of the AI and physics checks.

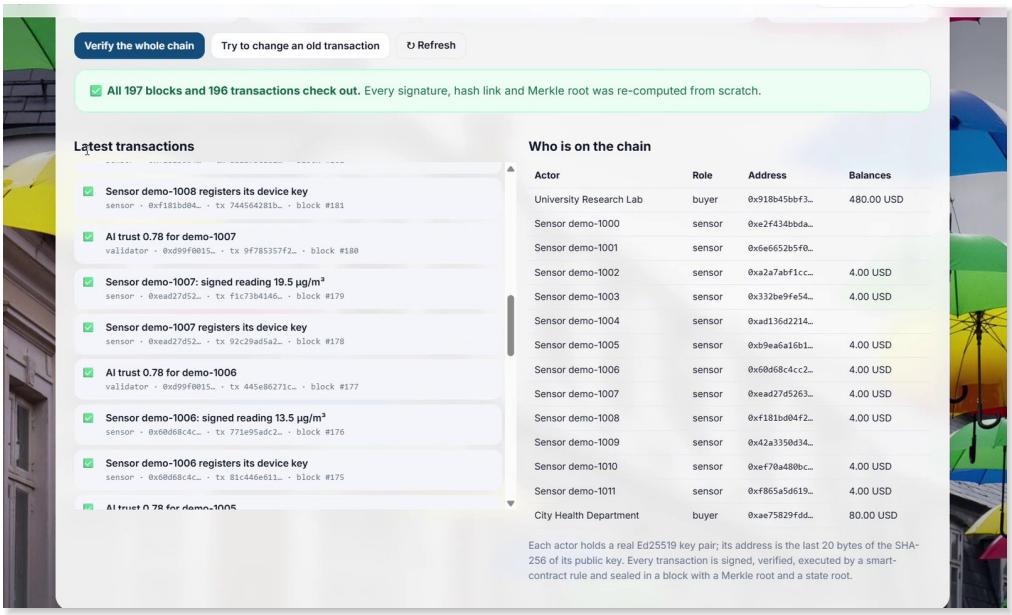


Figure 8: The ledger explorer: every signed transaction from every desk; full verification and the tamper test.

5 AI methods

Table 3: Methods, where they are used and why. All run in about a second on a laptop (pure NumPy/SciPy).

Method	Where	Purpose
Masked-feature reconstruction	Track 1, climate check	learn how signals co-move from unlabeled normal data; residuals flag tampering
Spatial masking + Beta trust	Track 4	predict a hidden sensor from its neighbours; Bayesian reputation drives revenue
Contrastive learning (InfoNCE)	Track 4	label-free embedding of 24-h windows → pollution regimes, unusual days
Metropolis–Hastings MCMC	Track 1	true emissions with a credible interval from three imperfect meters
Markov chain + Monte-Carlo	Tracks 2, 3, climate check	forecast uncertainty, plausibility of lifecycle paths, regime surprise
GEV / GPD extremes, BLUE fusion	climate check	return periods; optimal combination of two weather records [15]
ReAct agents + critic + judge	climate check	plan tools by value of information; try to refute; Bayes vs. Γ-minimax decision [16]
Fact-checked LLM (optional)	explanations, chat	friendlier text; every number must appear in the data, a second pass looks for unsupported claims

The contrastive encoder maps a window x to $z = \text{normalize}(Wx)$ and minimises, for two augmented views (u_i, v_i) of each window,

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(u_i^\top v_i/\tau)}{\sum_{j=1}^{N} \exp(u_i^\top v_j/\tau)},$$

with analytic gradients ($\tau = 0.2$). Sensor trust is the posterior mean of $\text{Beta}(2+a,\, 2+d)$ given agreement weight a and disagreement weight d. The emissions posterior combines the CEMS monitor ($\pm 5\,\%$), power output ($\pm 4\,\%$) and the fuel meter ($\pm 2\,\%$ with an unknown bias on flagged days); the Metropolis–Hastings acceptance rate was 76 % in the Berlin run.

6 Built for everyone

All labels are human-translated into 13 languages (English, Türkçe, Persian, Hindi, Chinese, French, Spanish, Arabic, Russian, German, Portuguese, Japanese, Uzbek) with matching fonts; Persian and Arabic switch the page to right-to-left. Generated text is translated on the fly by the browser's on-device translator or the local LLM, and a translation that changes any number is rejected. Every time zone works, including half-hour zones (Tehran +3:30, India +5:30), and any period can end today: the newest days come from the latest model analyses, bias-matched to ERA5 on 30 overlapping days.

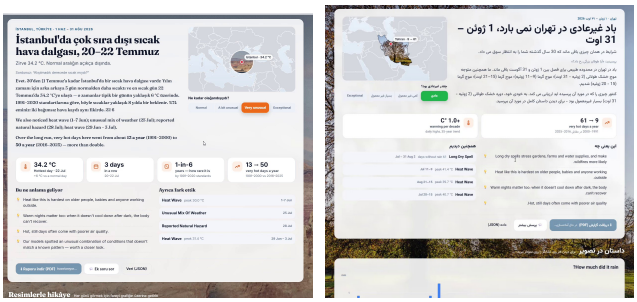


Figure 9: Istanbul in Turkish; Tehran in Persian (right-to-left).

Table 4: Open data sources (all keyless).

Source	Used for	Licence
Copernicus ERA5 & forecasts via Open-Meteo [11]	daily/hourly weather since 1940, forecasts, seasonal outlook	CC BY 4.0
NASA POWER	independent second weather record	public
Copernicus CAMS via Open-Meteo	air quality ($PM_{2.5}$, PM_{10}, NO_2, O_3)	CC BY 4.0
Climate TRACE [8]	asset-level emissions	CC BY 4.0
Energy-Charts, Fraunhofer ISE [9]	day-ahead electricity prices	CC BY 4.0
Open Food Facts + Agribalyse [10]	products, packaging, life-cycle CO_2	ODbL
Sensor.Community	live citizen PM sensors	ODbL
NASA GIBS & EONET, NOAA, GFZ, GDELT	satellite tiles, hazards, ENSO, Kp, news context	public

7 Evaluation and responsible AI

Testing. 25 automated tests cover Merkle proofs, signatures and tamper detection, every double-counting rule, all four desks offline, half-hour time zones, bias matching of recent days, text repair of product data and completeness of the translation table. API-shaped mocks exercise every live connector, and end-to-end browser tests drive the page in several languages.

Robustness. A failing connector never breaks a desk: the run falls back to the labelled demo world, the reason is shown, and each desk lists which data connections worked. LLM output is never trusted blindly; untrusted news and social text is injection-guarded and aggregate-only, and no personal data is stored.

8 Judging criteria

Criterion	How Dunyo addresses it
Climate impact	checkable heat, rain and air findings for cities; under-reported emissions caught; credits verified against physics and protected from double counting; honest hourly solar claims; traceable product footprints and recycling; rewarded, verified citizen data.
Innovation	one ledger for all four tracks; live "try to cheat" demonstrations; AI that doubts itself; self-supervised, contrastive and sampling methods each where they fit; Merkle provenance of an entire AI investigation.
Technical execution	14 open-data connectors with caching and fallbacks; Ed25519, Merkle and state roots; five Solidity contracts; 25 tests; every desk runs in seconds on a laptop.
Practical usefulness	any place on Earth, no keys, one command; PDF and JSON reports; useful to cities, NGOs, journalists, carbon buyers, energy communities, brands, recyclers, researchers and citizens.
Presentation	a news-style page with real photos, plain-language headlines and explained charts, 13 languages, a 4-minute narrated demo.

What's next. Deploy the contracts on a public testnet with multi-party attestations; connect real MRV meters and registries; add product categories as EU passports roll out; host Dunyo for city portals; open the data market to research groups.

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